MATHS MANTHAN
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Subject: Mathematics | **CBSE CLASS 10 – PERIODIC TEST 12** | **Time:** 60 Minutes
Class: Class X (CBSE) | **Chapter 13: Statistics** | **MM:** 30

Document Type: Official Step-by-Step Marking Scheme **Published for:** mathsmanthan.co.in **Test Paper:** Periodic Test 12

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, and final boxed answer with proper units.
2. In alternative (**OR**) questions, step marking applies equally to both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Alternative mathematically valid methods are fully credited.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Multiple Choice & Assertion-Reason MATHS MANTHAN | (6 × 1 = 6 Marks)

Q1. If the mean of $x, x+2, x+4, x+6, x+8$ is 24, then the value of x is: [1]

(A) 20 (B) 22 (C) 24 (D) 18

► **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A) 20

Justification:

$$\text{Mean} = \frac{x + (x+2) + (x+4) + (x+6) + (x+8)}{5} = \frac{5x + 20}{5} = x + 4$$

Given Mean $= 24 \implies x + 4 = 24 \implies \boxed{x = \mathbf{20}}$.

Q2. In a frequency distribution table, the frequency of the modal class is denoted by f_1. Then f_2 represents the: [1]

(A) frequency of the modal class
(B) frequency of the class succeeding the modal class
(C) frequency of the class preceding the modal class
(D) total frequency

► **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (B) frequency of the class succeeding the modal class

Justification: By convention in the standard CBSE mode formula, f_1 is the modal class frequency, f_0 is the preceding class frequency, and f_2 is the frequency of the succeeding class interval.

Q3. If $\sum f_i = 50, l = 60, cf = 15, f = 20$, and $h = 10$, the median of the distribution is: [1]

(A) 65 (B) 68 (C) 70 (D) 62.5

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (A) 65

Justification:

$$\text{Median} = l + \left(\frac{\frac{N}{2} - cf}{f}\right) \times h = 60 + \left(\frac{25 - 15}{20}\right) \times 10 = 60 + \left(\frac{10}{20}\right) \times 10 = 60 + 5 = \mathbf{65}.$$

$$\boxed{\text{Median} = \mathbf{65}}$$

Q4. For a moderately skewed frequency distribution, if the median is equal to its mean, then the mode of the distribution is: [1]

(A) equal to the mean (B) greater than the mean

(B) less than the mean (D) zero

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (A) equal to the mean

Justification: From empirical relationship:

$$\text{Mode} = 3\text{Median} - 2\text{Mean}$$

If Median = Mean = M, then Mode = $3M - 2M = M$ = Mean.

Q5. The mean of 100 observations is 50. If one observation 50 was replaced by 150, the resulting mean is: [1]

(A) 50.5 (B) 51.0 (C) 51.5 (D) 52.0

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (B) 51.0

Justification:

$$\text{Original Sum} = 100 \times 50 = 5000.$$

$$\text{New Sum} = 5000 - 50 + 150 = 5100.$$

$$\bar{x}_{\text{new}} = \frac{5100}{100} = \mathbf{51.0}.$$

$$\boxed{\text{Resulting Mean} = \mathbf{51.0}}$$

Q6. **Assertion (A):** If the median class is $20 - 30$ and the modal class is $30 - 40$, their respective class marks are 25 and 35.

Reason (R): Class mark is given by Class mark $= \dfrac{\text{Lower limit} + \text{Upper limit}}{2}$. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(C) (A) is true but (R) is false.

(D) (A) is false but (R) is true.

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (A)

Justification: Class mark of $20 - 30$ is $\frac{20+30}{2} = 25$, and for $30 - 40$ is $\frac{30+40}{2} = 35$. Both statements are true, and the formula given in Reason (R) is the exact mathematical rule that validates Assertion (A).

SECTION B: Very Short Answer Type — MATHS MANTHAN | (2 × 2 = 4 Marks)

Q7. Write the empirical relationship between the three measures of central tendency. Hence express Median in terms of Mean and Mode. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Empirical Relationship Declaration [1 Mark]

The empirical relationship between Mean, Median, and Mode is:

$$\text{Mode} = 3\text{Median} - 2\text{Mean}$$

Step 2: Expressing Median in terms of Mean and Mode [1 Mark]

$$3\text{Median} = \text{Mode} + 2\text{Mean} \implies \text{Median} = \frac{\text{Mode} + 2\text{Mean}}{3}.$$

$$\boxed{\text{Median} = \frac{\text{Mode} + \mathbf{2}\text{Mean}}{\mathbf{3}}}$$

Q8. For the following data, write the modal class and find its class width: [2]

Class	5 – 10	10 – 15	15 – 20	20 – 25	25 – 30	30 – 35
Frequency	6	11	21	23	14	5

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Identifying the Modal Class [1 Mark]

The maximum frequency is 23, which lies in the class interval **20 – 25**.

$\therefore$ Modal Class = **20 – 25**.

Step 2: Calculating Class Width [1 Mark]

$$\text{Class Width } (h) = \text{Upper Limit} - \text{Lower Limit} = 25 - 20 = \mathbf{5}.$$

$$\boxed{\text{Modal Class} = \mathbf{20 - 25}, \quad \text{Class Width} = \mathbf{5}}$$

SECTION C: Short Answer Type — MATHS MANTHAN | (2 × 3 = 6 Marks)

Q9. Find the mode of the following distribution: [3]

Class Interval	0 – 10	10 – 20	20 – 30	30 – 40	40 – 50
Frequency	8	12	10	11	9

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Step 1: Modal Class & Parameter Identification [1 Mark]

The maximum frequency is 12, which belongs to the class $\mathbf{10-20}$.
$\therefore$ Modal Class $= 10-20$.
Lower limit $(l) = 10$, class width $(h) = 10$.
$f_1 = 12$ (modal frequency), $f_0 = 8$ (preceding frequency), $f_2 = 10$ (succeeding frequency).

Step 2: Mode Formula & Substitution [1 Mark]

$$\text{Mode} = l + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2}\right) \times h$$

$$\text{Mode} = 10 + \left(\frac{12-8}{2(12)-8-10}\right) \times 10 = 10 + \left(\frac{4}{24-18}\right) \times 10 = 10 + \left(\frac{4}{6}\right) \times 10$$

Step 3: Calculation & Final Boxed Answer [1 Mark]

$$\text{Mode} = 10 + \frac{20}{3} = 10 + 6.667 \approx \mathbf{16.67} \quad \left(\text{or } 16\frac{2}{3}\right)$$

$$\boxed{\text{Mode} = \mathbf{16.67} \quad \left(\text{or } \mathbf{16\frac{2}{3}}\right)}$$

Q10. Find the median of the following distribution of scores: [3]

Class	100 – 110	110 – 120	120 – 130	130 – 140	140 – 150
Frequency	6	10	15	13	6

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Step 1: Cumulative Frequency Table [1 Mark]

Class	**Frequency (f_i)**	**Cumulative Frequency (cf)**
$100-110$	6	6
$110-120$	10	$6+10=16$
$120-130$	15	$16+15=31$
$130-140$	13	$31+13=44$
$140-150$	6	$44+6=50$
Total	$N = \sum f_i = 50$	

Step 2: Median Class & Parameters [1 Mark]

$N = 50 \implies \frac{N}{2} = 25$.
Cumulative frequency just greater than 25 is 31, which corresponds to class $\mathbf{120-130}$.
$\therefore$ Median Class $= 120-130$.
Lower limit $(l) = 120$, $cf = 16$ (preceding cf), frequency $(f) = 15$, class width $(h) = 10$.

Step 3: Median Formula & Calculation [1 Mark]

$$\text{Median} = l + \left(\frac{\frac{N}{2} - cf}{f}\right) \times h = 120 + \left(\frac{25-16}{15}\right) \times 10 = 120 + \left(\frac{9}{15}\right) \times 10 = 120 + 6 = \mathbf{126}.$$

$$\boxed{\text{Median Score} = \mathbf{126}}$$

SECTION D: Long Answer Type

MATHS MANTHAN | (2 × 5 = 10 Marks)

Q11. An incomplete distribution having total frequency 230 has median 46. Find the missing frequencies f_1 and f_2: **[5]**

Class Interval	$10-20$	$20-30$	$30-40$	$40-50$	$50-60$	$60-70$	$70-80$
Frequency	12	30	f_1	65	f_2	25	18

► **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]

Step 1: Total Frequency Equation [1.5 Marks]

$$\sum f_i = 12 + 30 + f_1 + 65 + f_2 + 25 + 18 = 150 + f_1 + f_2 = 230$$

$$f_1 + f_2 = 230 - 150 \implies f_1 + f_2 = 80 \quad \text{— (i)}$$

Step 2: Cumulative Frequency Table [1 Mark]

Class Interval	Frequency (f_i)	Cumulative Frequency (cf)
$10-20$	12	12
$20-30$	30	42
$30-40$	f_1	$42 + f_1$
$40-50$	65	$107 + f_1$
$50-60$	f_2	$107 + f_1 + f_2$
$60-70$	25	$132 + f_1 + f_2$
$70-80$	18	$150 + f_1 + f_2$

Step 3: Median Class Parameters & Solving for f_1 [1.5 Marks]

Given Median $= 46$, which lies in the class interval **$40-50$**.

$\therefore$ Median Class $= 40-50$.

$l = 40, h = 10, f = 65, \dfrac{N}{2} = \dfrac{230}{2} = 115, cf = 42 + f_1$ (preceding cf).

$$\text{Median} = l + \left(\frac{\frac{N}{2} - cf}{f}\right) \times h \implies 46 = 40 + \left(\frac{115 - (42 + f_1)}{65}\right) \times 10$$

$$6 = \left(\frac{73 - f_1}{65}\right) \times 10 \implies 6 \times 65 = 10(73 - f_1) \implies 390 = 730 - 10f_1$$

$$10f_1 = 730 - 390 = 340 \implies f_1 = \mathbf{34}.$$

Step 4: Finding f_2 & Final Result [1 Mark]

From Equation (i):

$$34 + f_2 = 80 \implies f_2 = 80 - 34 = \mathbf{46}.$$

$$\boxed{f_1 = \mathbf{34}, \quad f_2 = \mathbf{46}}$$

Q12. In a hospital, the ages of 360 patients admitted during a month were recorded. Find both the mean age (using step-deviation method) and modal age: **[5]**

Age (in years)	$5-15$	$15-25$	$25-35$	$35-45$	$45-55$	$55-65$
Patients	30	60	100	80	50	40

► **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]

Part 1: Mean Age by Step-Deviation Method ($a = 30, h = 10$) [2.5 Marks]

Age (years)	Patients (f_i)	Class Mark (x_i)	$u_i = \frac{x_i - 30}{10}$	$f_i u_i$
$5 - 15$	30	10	-2	-60
$15 - 25$	60	20	-1	-60
$25 - 35$	100	30	0	0
$35 - 45$	80	40	$+1$	$+80$
$45 - 55$	50	50	$+2$	$+100$
$55 - 65$	40	60	$+3$	$+120$
Total	$\sum f_i = 360$			$\sum f_i u_i = 180$

$$\bar{x} = a + \left(\frac{\sum f_i u_i}{\sum f_i}\right) \times h = 30 + \left(\frac{180}{360}\right) \times 10 = 30 + 5 = \mathbf{35} \text{ years.}$$

Part 2: Modal Age Calculation [2.5 Marks]

The maximum frequency is 100, which lies in the modal class $\mathbf{25 - 35}$.
$l = 25, h = 10, f_1 = 100, f_0 = 60, f_2 = 80$.

$$\text{Mode} = l + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2}\right) \times h = 25 + \left(\frac{100 - 60}{2(100) - 60 - 80}\right) \times 10 = 25 + \left(\frac{40}{60}\right) \times 10$$

$$\text{Mode} = 25 + \frac{20}{3} = 25 + 6.667 \approx \mathbf{31.67} \text{ years} \quad \left(\text{or } 31\frac{2}{3} \text{ years}\right).$$

Mean Age = **35** years, Modal Age = **31.67** years $\left(\text{or } \mathbf{31\frac{2}{3}} \text{ years}\right)$

SECTION E: Case Study Based Question

MATHS MANTHAN | (1 × 4 = 4 Marks)

Q13. CASE STUDY: STATE POLLUTION CONTROL BOARD CITY AMBIENT AIR QUALITY INDEX (AQI) [4 Marks]

During winter inversion months, environmental scientists recorded the 24-hour peak Air Quality Index (AQI) values across 70 continuous ambient air monitoring stations in a metropolitan region.

AQI Range	$50 - 100$	$100 - 150$	$150 - 200$	$200 - 250$	$250 - 300$	$300 - 350$
Monitoring Stations	5	12	20	18	10	5

(i) Calculate the mode of the Air Quality Index (AQI) recorded across stations. [2]

(ii) Find the median class for the ambient air quality readings. [1]

(iii) How many monitoring stations recorded severe or very poor air (AQI ≥ 200)? [1]

► **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 4 Marks]

Subpart (i): Modal AQI Calculation [2 Marks]

The maximum number of monitoring stations is 20, corresponding to modal class **150 – 200**.
$l = 150, h = 50, f_1 = 20, f_0 = 12, f_2 = 18$.

$$\text{Mode} = l + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2}\right) \times h \text{[1 Mark]}$$

$$\text{Mode} = 150 + \left(\frac{20 - 12}{2(20) - 12 - 18}\right) \times 50 = 150 + \left(\frac{8}{40 - 30}\right) \times 50 = 150 + 40 = \mathbf{190} \text{[1 Mark]}$$

$$\boxed{\text{Mode of AQI} = \mathbf{190}}$$

Subpart (ii): Median Class Identification [1 Mark]

$N = 70 \implies \dfrac{N}{2} = 35$.
Cumulative frequencies: $50 - 100 : 5$; $100 - 150 : 17$; $150 - 200 : 37$; $200 - 250 : 55$; $250 - 300 : 65$; $300 - 350 : 70$.
The cumulative frequency just greater than 35 is 37, corresponding to median class **150 – 200**.

$$\boxed{\text{Median Class} = \mathbf{150 - 200}}$$

Subpart (iii): Stations with AQI ≥ 200 [1 Mark]

Monitoring stations in intervals $[200 - 250]$, $[250 - 300]$, and $[300 - 350]$:

$$\text{Total Stations} = 18 + 10 + 5 = \mathbf{33}.$$

$$\boxed{\text{Monitoring Stations with AQI } \geq 200 = \mathbf{33}}$$

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